

Solve problems involving the volume and surface area of right prisms and cylinders using appropriate units

Learning intention

- solve problems involving the volume and surface area of right prisms and cylinders using appropriate units.

Teach and model

- finding different prisms that have the same volume but different surface areas.
- analysing nets of objects to generate short cuts and establish formulas for surface area.
- determining the amount of material needed to make can-coolers for a class fundraising project and working out the most cost-efficient way to cut out the pieces.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.